

Manufacturing Line Downtime Analysis & Capacity Planning

Project Report | October 2025

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Status: Ready for Implementation

Executive Summary

Manufacturing Line Downtime Analysis dashboard combines downtime analysis, predictive forecasting, and operational capacity planning into a unified Power BI solution. Delivers 70% reporting time reduction and 15-20% capacity improvement within 4 weeks.

Key Deliverables: Interactive Dashboard | Forecasting Model | Capacity Planning | Comprehensive Report

1. Project Objectives

- 1.1 Analyze Downtime** - Classify and measure downtime by factor, product, operator, and time period
- 1.2 Performance Benchmarking** - Compare actual vs. minimum batch times
- 1.3 Forecast Downtime** - Predict next-day downtime for operational planning
- 1.4 Plan Capacity** - Calculate feasible batch production: $Batches = (Available\ Time - Predicted\ Downtime) \div Min\ Batch\ Time$
- 1.5 Support Decisions** - Provide actionable insights through dashboards and reports

2. Business Impact

Metric	Target	Impact
Reporting Efficiency	70% reduction	4 hrs/week → <1.5 hrs/week
Decision Speed	Same-day insights	vs. 48-hour delays
Capacity Improvement	+15-20% throughput	Increased batch production
Downtime Reduction	15-20% lower incidents	Process improvements
User Adoption	>80% within 30 days	Strong engagement

3. Team Structure

Member	Role	Responsibilities
Mohamed Mahmoud Sayed Rashwan	Team Leader	Data model design, key measures, visualizations, stakeholder alignment
Hassan Hany Hassan Shohaib	Data & ETL Engineer	Data cleaning, unpivot transformations, relationships, data quality
Mohamed Fathi Hamed Elnagar	Forecast & Capacity Analyst	Forecasting logic, batch calculations, predictive indicators
Abdelrahman Salaheldien Youssef Saraya	QA & Deployment	Performance testing, UAT, distribution, feedback

4. Project Timeline

Week 1: Data Preparation

Clean data, transform downtime table (unpivot), build relationships → **Deliverable:** Analysis-ready dataset

Week 2: Analysis & Forecasting

Define analytical questions, build forecast model, develop batch calculations → **Deliverable:** Forecasting logic & formulas

Week 3: Dashboard Development

Create visualizations, design KPI cards, add filters → **Deliverable:** Complete Power BI dashboard

Week 4: Final Report & Deployment

Prepare report, create presentations, test distribution → **Deliverable:** Project report, presentation, adoption plan

5. Technology Stack

Component	Tool	Purpose
BI Platform	Power BI Desktop & Service	Dashboard & reports
Data Processing	Power Query	ETL transformations
Calculations	DAX	Measures & forecasting
Data Source	Manufacturing_Line_Productivity.xlsx	Production data
Reference	data_dictionary.csv	Field definitions

6. Data & Transformations

Core Tables:

Line Productivity (batch details) | Products (characteristics) | Line Downtime (downtime events - wide format) | Downtime Factors (definitions)

Key Transformations:

Unpivot downtime table (wide → long) | Calculate batch duration | Link tables via keys | Aggregate downtime dimensions

7. Batch Capacity Formula

Batches = (Available Time - Predicted Downtime) ÷ Min Batch Time

Example: Available Time: 480 min | Predicted Downtime: 45 min | Min Batch Time: 30 min

Result: $(480 - 45) \div 30 = 14.5 \rightarrow$ **14 batches**

DAX: $\text{INT}((480 - [\text{Next Day Downtime Forecast}]) / [\text{Min Batch Time}])$

8. Dashboard Components

KPI Cards: Total Downtime | Average Batch Duration | Next-Day Forecast | Feasible Batch Count | Performance vs Target

Visualizations: Top 5 Downtime Factors (bar chart) | Downtime Trend (line chart) | Downtime by Product (column) | Downtime by Operator (table) | Product × Operator Matrix

Filters: Date range | Product multi-select | Operator multi-select | Factor drill-down | Cross-filtering enabled

9. Key Performance Indicators

Data Quality: Completeness >95% | Validity >98% | Error Rate <2%

Performance: Load Time <5 sec | Refresh Time <10 min | Query Response <2 sec

Business Impact: Usability >4.0/5.0 | Adoption >80% | Report Access >90% | Insights >5 monthly | Capacity +15-20%

10. Implementation Readiness

Prerequisites Met:

- ✓ Team assigned with clear roles
- ✓ Data sources identified
- ✓ Technical architecture designed

- ✓ Success criteria established
- ✓ Timeline defined (4 weeks)

Expected Outcomes:

Interactive Dashboard | Next-day Forecasting | Automated Batch Calculations | Actionable Insights |
>80% User Engagement

Status: ✓ READY FOR IMPLEMENTATION | Timeline: 4 weeks | Expected ROI: 70% efficiency + 15-20% capacity improvement

Prepared by: Mohamed Mahmoud Sayed Rashwan | **Date:** October 2025 | **Version:** 1.0